Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Seonghwan (Sam) Kim and Keekyoung Kim	Email	sskim@ucalgary.ca
Institute Name	University of Toronto			
Address	Department of Mechanical & Industrial Engineering, 5 King's College Rd. Toronto, ON, Canada M5S 3G8			

2. Technology Overview

Title of Proposed Technology	Development of an AI-Enabled Physical Sensor System for Biological Wastewater Processes			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (✓), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (), Cybersecurity ( ), AI (✓), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical Electronics		Mechanical Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Measurement Sensors and Components	Energy/Environment Mechanical Systems	Energy/Environmental Control Equipment
Concept of Proposed Technology	This proposed technology aims to develop an Artificial Intelligence (AI)-enabled, physical sensor–system for biological wastewater treatment processes by integrating advanced Metal-Organic Framework (MOF) composite or Carbon Quantum Dots (CQDs) with Graphitic Carbon Nitride (gC_3N_4) composite-based and optical sensing technologies with high-resolution multivariate signal analysis and AI-based interpretation models. The goal is to overcome the limitations of conventional laboratory-dependent water quality monitoring and establish a continuous, high-reliability platform suitable for field deployment and future commercialization. Conventional water quality indicators such as Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Organic Carbon (TOC), Suspended Solids (SS), Total Nitrogen (T-N), and Total Phosphorus (T-P) depend on intermittent sampling and laboratory analysis, which restrict real-time process visibility and delay operational decision-making. To address these limitations, the proposed technology employs a high-sensitivity physical and optical sensing platform that continuously measures water-quality-responsive signals, including electrical conductivity, turbidity,			

	Oxidation-Reduction Potential (ORP), temperature, UV–Vis absorbance, fluorescence signatures, and MOF or or CQDs with gC_3N_4 composite-based optical interactions. The MOF or CQDs with gC_3N_4 composite materials provide enhanced selectivity, stability, and molecular-level detection capabilities, enabling more detailed characterization of water quality dynamics compared to traditional sensors. The multivariate time-series data generated from these sensors will be used to train and operate AI-based virtual sensor models capable of real-time estimation of key water quality indicators that are difficult or impossible to measure continuously. By learning nonlinear correlations and temporal structures within the physical and optical signals, the AI virtual sensor predicts parameters such as BOD, COD, TOC, SS, T-N, and T-P with high accuracy and minimal maintenance requirements. Data-driven drift compensation and long-term signal stabilization techniques will further ensure robustness under real wastewater treatment conditions. In addition, the proposed technology establishes a foundation for predictive process analysis and digital twin integration, enabling early detection of process disturbances, influent variability forecasting, and optimization of operational strategies such as aeration control and nutrient removal. This transforms the system from a simple measurement device into an intelligent monitoring and decision-support platform for biological wastewater treatment. By combining the advanced materials–based sensor technologies with AI-enabled multivariate signal interpretation, and by collaborating with Small and Medium-sized Enterprises (SMEs) for system integration, field deployment, and commercialization, this project aims to deliver a next-generation real-time monitoring and virtual sensing solution that supports continuous, reagent-free operation, enhances treatment stability, and significantly improves environmental process management capabilities.
Rationale for Joint R&D Project	The proposed technology combines advanced MOF or CQDs with gC_3N_4 composite-based and optical sensor technologies with AI-enabled virtual sensing and process prediction approaches to create a next-generation real-time monitoring platform for biological wastewater treatment. While the core sensing and analytical models will be developed by the global research institute, successful deployment, validation, and commercialization of this technology require close collaboration with SMEs that possess practical experience in wastewater treatment operations, industrial system engineering, and field integration. To transition from laboratory-scale sensor innovations to a fully functional, field-operational monitoring system, comprehensive activities such as hardware packaging, system integration, on-site testing, long-term durability evaluation, and compliance with industrial safety standards must be conducted. SMEs play a critical role in these aspects, as they understand domestic operational environments, regulatory requirements, and market needs. Their involvement ensures that the developed system can be manufactured, installed, and maintained in real wastewater treatment facilities under realistic operational constraints. Another essential component of this collaboration is the establishment of a data-driven operational framework, including virtual sensing and digital-twin-based process analysis. Although the research institute will develop sensing technologies and analytical methodologies, field verification using real wastewater conditions in actual environment is necessary to refine the system, validate prediction accuracy, and confirm stability under complex biological treatment dynamics. SMEs can provide access to demonstration sites, perform long-term pilot operation, and evaluate system performance under actual influent variability, environmental fluctuations, and maintenance conditions. Furthermore, joint R&D enables both parties to integrate advanced sensing materials with industrial-grade monitoring systems. SMEs can lead efforts in enclosure design, instrumentation, communication interfaces, manufacturing scalability assessment, and certification processes required for domestic and international deployment. This collaboration significantly accelerates technology transfer and strengthens commercialization potential. Through this international partnership, the project ensures that the advanced materials–based sensing technology is fully adapted to practical field environments, while SMEs secure a

	differentiated, high-value technology with strong competitiveness in the smart water and environmental monitoring markets. The collaboration ultimately provides a comprehensive pathway from laboratory innovation to industrial application, establishing a robust foundation for global commercialization of next-generation wastewater monitoring and virtual sensing systems.
Key Words	MOF or CQDs with gC_3N_4 composite-based Optical Sensor AI-enabled Virtual Sensor Real-time Water Quality Monitoring Digital Twin for Wastewater Treatment Data-driven Process Optimization

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (7-8)
R&D Objectives	The primary objective of this project is to develop an AI-enabled, MOF or CQDs with gC_3N_4 composite -based real-time optical sensing system capable of continuously estimating key water quality indicators in biological wastewater treatment processes. By integrating advanced sensing materials with multivariate time-series analysis and AI-based prediction models, the project aims to establish a stable, reagent-free, and field-deployable monitoring platform. Specific objectives are as follows: 1. Develop high-sensitivity physical and optical sensing elements, including MOF or CQDs with gC_3N_4 composite based optical sensors, that can operate reliably under biological wastewater conditions with fluctuations in temperature, pH, biomass concentration, and suspended solids. 2. Construct a continuous data acquisition system that captures multivariate signals such as electrical conductivity, ORP, UV–Vis absorbance, fluorescence, and MOF or CQDs with gC_3N_4 composite derived optical fingerprints at high temporal resolution suitable for AI model training. 3. Design and implement AI-based virtual sensor models capable of real-time estimation of key parameters including BOD, COD, TOC, SS, T-N, and T-P by learning temporal and nonlinear relationships among physical and optical signals. 4. Develop signal stability enhancement methods, including drift analysis, compensation algorithms, noise reduction, and long-term reliability assessment to ensure stable operation over extended periods without chemical reagents. 5. Integrate the sensing system, data processing modules, and AI models into a field-deployable monitoring unit and validate its performance through laboratory and pilot-scale wastewater tests that reflect representative biological treatment conditions. 6. Establish the foundation for predictive process analysis and digital twin compatibility by enabling the virtual sensor outputs to support process disturbance detection, influent variability forecasting, and operational optimization.			
R&D Target Specifications	The target specifications of this project are defined to ensure that the developed MOF or CQDs with gC_3N_4 composite-based optical sensing system can provide reliable, continuous, and accurate monitoring of biological wastewater treatment processes under realistic operating conditions. **1. Continuous Operation Period: ≥ 6 months** The system shall maintain stable performance without frequent sensor replacement or chemical reagent use, ensuring long-term applicability in wastewater environments. **2. Sampling Interval (Measurement Cycle): ≤ 10 minutes** High-frequency data acquisition is required to support real-time AI model inference and rapid detection of process fluctuations. **3. Response Time: ≤ 1 minute** The sensing system must capture rapid changes in water quality and support timely process assessment. **4. Measurement Accuracy: Within ±5% of the reading** Accuracy must be sufficient for biological wastewater monitoring and for training AI-based virtual sensor models. **5. Long-Term Signal Drift: ≤ ±5% per month** MOF-based and optical sensing elements shall maintain signal stability and reproducibility during extended immersion in wastewater. **6. System Availability: ≥ 95%**			

	The monitoring platform must sustain high operational uptime with minimal downtime or data loss. 7. **Operating Environmental Conditions:** Stable operation under 0–40°C, pH 5–9, high suspended solids, and continuous immersion in activated sludge or mixed liquor. 8. **AI Data Availability: Valid Data Ratio ≥ 99%** Data quality must be sufficient to ensure robust AI model training, prediction accuracy, and digital twin compatibility. 9. **System Connectivity and Integration:** Support for standard data interfaces (API or equivalent) enabling linkage with AI analysis modules, virtual sensor engines, and digital process models.
R&D Plan	The project will be carried out in four phases to advance MOF or CQDs with gC_3N_4 composite-based optical sensing system concepts into a field-deployable, AI-enabled sensing system suitable for biological wastewater treatment processes. **Phase 1: Sensor and Measurement System Optimization** 1. Optimize MOF or CQDs with gC_3N_4 composite-based optical sensing elements and supporting physicochemical sensors for long-term continuous operation in wastewater environments. 2. Improve sensor surface materials, coatings, and structural designs to enhance anti-fouling performance and durability against biofilm accumulation and suspended solids. 3. Analyze environmental factors such as pH, temperature variations, and biomass concentration to quantify their impact on sensor signals and develop mitigation strategies. 4. Define the overall measurement system architecture, including sensor configuration, power management, data acquisition pathways, and communication structure. **Phase 2: Data Processing Framework and AI Model Development** 1. Build a high-resolution multivariate time-series dataset through laboratory-scale tests simulating representative biological wastewater conditions. 2. Develop data preprocessing methods such as noise filtering, baseline correction, missing-value handling, and normalization to ensure AI compatibility. 3. Conduct signal drift analysis and develop data-driven drift compensation models to support long-term stability. 4. Design AI-based virtual sensor models capable of estimating key parameters such as BOD, COD, TOC, SS, T-N, and T-P using temporal and nonlinear relationships among physical and optical signals. 5. Establish standardized data interfaces to connect sensing modules with AI analysis and digital process modeling tools. **Phase 3: System Integration and Pilot-Scale Field Validation** 1. Integrate optimized sensing elements, data processing modules, and AI models into a unified prototype monitoring unit. 2. Install the prototype in pilot-scale biological wastewater treatment environments to conduct long-term continuous operation tests. 3. Evaluate performance based on accuracy, response time, drift behavior, stability, fault tolerance, and maintenance requirements. 4. Refine sensor design, AI models, and system architecture through iterative improvement based on field test results. **Phase 4: Commercial Readiness and Deployment Assessment**

	1. Assess manufacturability, cost structure, and scalability for transition from prototype to commercial-grade monitoring units. 2. Evaluate long-term operational robustness, cleaning requirements, and maintenance intervals under real wastewater treatment conditions. 3. Review regulatory, industrial safety, and certification requirements applicable to wastewater instrumentation and prepare compliance strategies. 4. Develop business-focused documentation, including technical specifications, application guidelines, and market entry strategies to support commercialization by SMEs.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will lead the development of core sensing technologies, data-processing methodologies, and AI-based virtual sensing models essential for achieving the project's technical objectives. The institute will provide the scientific foundation required to advance MOF or CQDs with gC_3N_4 composite-based optical sensor concepts into a reliable real-time monitoring system applicable to biological wastewater treatment. **1. Development and optimization of sensing technologies** Design, synthesize, and optimize MOF or CQDs with gC_3N_4 composite-based optical sensing elements and supporting physicochemical sensors suitable for long-term continuous operation in wastewater environments. Improve sensor materials, surface properties, and structural configurations to enhance signal stability, anti-fouling performance, and measurement reproducibility. **2. Environmental impact assessment and interference mitigation** Conduct systematic evaluation of environmental factors such as temperature fluctuations, pH variability, biomass accumulation, and suspended solids. Develop methods to reduce interference, quantify drift behavior, and ensure stable signal acquisition during extended immersion. **3. Data processing model development** Establish data preprocessing procedures including noise reduction, baseline correction, calibration strategies, and signal harmonization. Analyze long-term drift patterns and develop data-driven compensation algorithms to support stable AI performance. **4. AI-based virtual sensor model design** Construct AI models capable of estimating key water quality indicators by learning correlations within multivariate time-series signals from the sensing platform. Validate model accuracy, robustness, and responsiveness under laboratory and pilot-scale wastewater conditions. **5. Laboratory-scale and pilot-scale performance evaluation** Conduct controlled laboratory experiments and pilot-scale tests to verify accuracy, response time, stability, and durability of the sensing system. Assess long-term performance under conditions representative of biological wastewater treatment processes. **6. Technical documentation and technology transfer support** Prepare standard operating procedures, calibration guidelines, test protocols, and sensor performance specifications. Provide technical materials and support necessary for smooth technology transfer, system integration, and commercialization activities conducted by SMEs.
Expected R&R of Korean SME Partner	The SME partner will be responsible for translating the research outcomes into a field-deployable, industrial-grade monitoring system. The SME will lead system integration, pilot-site operation, manufacturability assessment, and commercialization activities necessary to ensure successful deployment in wastewater treatment facilities. **1. Definition of industrial requirements and system specifications** Identify practical operational needs, regulatory considerations, maintenance constraints, and environmental conditions typical of domestic wastewater treatment facilities. Translate these requirements into concrete system specifications, including durability, installation methods,

	sensor protection structures, power management, and communication interfaces. 2. **System integration, packaging, and engineering design** Integrate MOF or CQDs with gC_3N_4 composite-based optical sensor modules with industrial-grade data acquisition hardware, enclosures, wiring, protective housings, and environmental sealing. Develop mechanical designs that ensure resistance to fouling, corrosion, vibration, and long-term immersion in biological wastewater conditions. 3. **Field installation and long-term operational testing** Deploy the prototype system in actual wastewater treatment facilities. Conduct long-duration performance evaluations including stability, accuracy validation, environmental stress testing, cleaning requirements, failure modes, and maintenance interval assessment under real influent and operational variability. 5. **Feedback for iterative improvement** Provide field-based feedback to refine sensor designs, system architecture, data acquisition reliability, and overall usability. Collaborate with the research institute to adjust system elements based on operational challenges encountered in the field. 4. **Manufacturing process development and scalability evaluation** Develop practical pathways for mass production, including component sourcing, assembly procedures, quality assurance steps, and cost optimization. Assess scalability and prepare the foundation for transitioning from prototype units to commercial products. 6. **Certification, regulatory compliance, and market entry preparation** Review and apply relevant Korean industrial safety standards, water-quality instrumentation certification requirements, and environmental equipment regulations. Prepare documentation and procedures required for regulatory approval and market introduction. 5. **Commercialization planning and business development** Develop commercialization strategies, including product packaging, service models, customer engagement, and pilot deployment pathways. Support the establishment of market channels for domestic and international distribution of the monitoring system.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This international joint R&D project aims to deliver a next-generation real-time monitoring and sensing system that significantly improves the operation, reliability, and efficiency of biological wastewater treatment processes. By combining the advanced MOF or CQDs with gC_3N_4 composite and optical sensor technologies with AI-based multivariate interpretation and the field integration capabilities of SMEs, the project is expected to produce meaningful technical, industrial, and societal impacts. **Technical Outcomes** 1. **Establishment of a continuous, reagent-free monitoring platform** The project will demonstrate a sensing system capable of stable operation for more than 12 months without chemical reagents, enabling reliable, long-term acquisition of multivariate signals in biological wastewater environments. 2. **High-accuracy sensing of key water quality parameters** AI-based models trained on MOF or CQDs with gC_3N_4 composite and optical-derived signals will provide accurate real-time estimation of BOD, COD, TOC, SS, T-N, and T-P, reducing dependence on laboratory analysis and improving responsiveness to process fluctuations. 3. **Enhanced process visibility and early detection of disturbances** Continuous, high-resolution sensor data combined with virtual sensing outputs will support early identification of abnormal trends, influent variability, and operational instability. 4. **Compatibility with predictive and digital process analysis** The system will generate data suitable for integration with digital twin frameworks, enabling predictive modeling, scenario simulation, and data-driven operational optimization.

Industrial and Commercial Impact

1. **Acceleration of technology transfer and commercialization**
   Collaboration with SMEs ensures practical system packaging, manufacturability assessment, and compliance with domestic certification standards, facilitating transition from laboratory innovation to commercial products.
2. **Competitive advantage for SMEs in the environmental monitoring market**
   Access to advanced sensing materials and AI-based sensor technologies will provide differentiated capabilities and enhance competitiveness in domestic and global smart water industries.
3. **Reduction of operational and maintenance costs**
   Reagent-free sensing, longer maintenance intervals, and automated virtual analysis reduce labor demands, consumable costs, and unexpected downtime in wastewater treatment facilities.
4. **Improved effluent quality compliance and operational stability**
   Real-time insights into water quality trends support more stable process control, enabling facilities to consistently meet regulatory discharge standards.

Societal and Environmental Impact

1. **Contribution to sustainable water management**
   More efficient, data-driven wastewater operation reduces energy use, enhances treatment efficiency, and supports broader sustainability and ESG objectives.
2. **Reduction of environmental footprint**
   Eliminating chemical reagents minimizes hazardous waste generation and decreases environmental impact associated with traditional laboratory-based monitoring.
3. **Strengthened public health and environmental protection**
   Improved monitoring accuracy and faster response to treatment disturbances help ensure safer effluent discharge and better protection of receiving water bodies.

6. References

(1) Vanrolleghem, P. A., & Lee, D. S. (2003), On-line monitoring equipment for wastewater treatment processes: state of the art, Water Science and Technology, 47(2), 1–34.

(2) Haimi, H., Mulas, M., Corona, F., & Vahala, R. (2013), Data-derived soft-sensors for biological wastewater treatment plants: an overview, Environmental Modelling & Software, 47, 88–107.

(3) Olsson, G., Nielsen, M. K., Yuan, Z., Lynggaard-Jensen, A., & Steyer, J. P. (2014), Instrumentation, control and automation in wastewater systems, IWA Publishing, London.

(4) Jeppsson, U., & Olsson, G. (2004), Control of wastewater treatment processes, IWA Publishing, London.

(5) Yuan, Z., Keller, J., & Batstone, D. J. (2012), The role of sensors and control in wastewater treatment, Water Research, 46(15), 4845–4862.

(6) Corominas, L., Flores-Alsina, X., Snip, L., & Vanrolleghem, P. A. (2011), Comparison of different modeling approaches to better evaluate greenhouse gas emissions from whole wastewater treatment plants, Biotechnology and Bioengineering, 108(12), 2854–2863.

(7) García, L., Barreiro-Gomez, J., Escobar, E., Téllez, D., Quijano, N., & Ocampo-Martinez, C. (2015), Modeling and real-time control of wastewater treatment plants: A review, Journal of Process Control, 27, 1–14.

(8) Sin, G., Van Hulle, S. W. H., De Pauw, D. J. W., van Griensven, A., & Vanrolleghem, P. A. (2005), A critical comparison of systematic calibration protocols for activated sludge models, Biotechnology and Bioengineering, 91(4), 455–478.

(9) ISO 15839:2003, Water quality — On-line sensors/analyzing equipment for water — Specifications and performance tests.

(10) U.S. Environmental Protection Agency (EPA) (2013), Wastewater Technology Fact Sheet: Sensors and Instrumentation, United States Environmental Protection Agency.

(11) Zandi, P., Phani, A., & Kim, S. (2025). Detecting Cr6+ at≈ 100 pM Concentration with Fluorescence Enhancement Signatures in a Novel Eco-Fluorophore: Matching WHO's 96 pM Recommended Standard for Drinking Water. *Advanced Materials*, 2504142.

(12) Zandi, P., Phani, A., & Kim, S. (2024). High-Performance P-N Junction Heterostructure with Carbon Quantum Dot and Nitrogen Self-Doped Graphitic Carbon Nitride for Visible Light Photodetection. *Advanced Optical Materials*, *12*(29), 2401317.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization